Reading a Rational Graph from Its Formula

Simplifying to find holes, locating vertical asymptotes, finding intercepts, and reading end behaviour for a horizontal asymptote

Before you start. Four questions answer almost everything about a rational graph, and the order matters.

- **Factor first.** A factor that cancels leaves a *hole*; a factor still in the denominator after cancelling is a *vertical asymptote*.
- **Intercepts.** The x-intercepts are the zeros of the *simplified* numerator; the y-intercept is the value at $x = 0$.
- **End behaviour.** Compare the degrees top and bottom to decide the horizontal asymptote.
- Show the work that produced each feature — a bare list of answers earns nothing here.

1. Factor the numerator and the denominator completely, then write f as a single factored fraction.

$$f(x) = \frac{x^2 - 9}{x^2 + 2x - 15}$$

Answer: ________________

2. Let $g(x) = \dfrac{x^2 - x - 6}{x^2 - 4}$.

(a) Write g in lowest terms. (b) The graph of g has a hole. Give its coordinates.

Answer: ________________

3. Find every vertical asymptote of $h(x) = \dfrac{x+1}{x^2 - 3x - 10}$. Show why nothing cancels first.

Answer: ____________________

4. For $p(x) = \dfrac{x^2 - 2x - 8}{x^2 + 4x + 3}$, find
(a) every x-intercept, (b) the y-intercept.

Answer: ____________________

5. Find the horizontal asymptote of $q(x) = \dfrac{3x^2 - 5}{x^2 + 4}$ by describing what happens to $q(x)$ as x grows without bound.

Answer: ____________________

6. Dana was asked for the intercepts of $r(x) = \dfrac{2x-6}{x^2-x-12}$. She wrote: "$r(0) = -0.5$, so the y-intercept is $(0, -0.5)$." Find and fix her mistake, then give the x-intercept as well.

Answer: ________________

7. $s(x) = \dfrac{x+4}{x^2-16}$ is undefined at two values of x. Find both, decide which one is a vertical asymptote and which one is a hole, and give the coordinates of the hole.

Answer: ________________

8. Find the horizontal asymptote of $t(x) = \dfrac{2x+7}{x^2+3x}$, and say in one sentence what the graph does far out to the right.

Answer: ________________

9. Both $x = 2$ and $x = -1$ make the numerator of $u(x) = \dfrac{x^2 - x - 2}{x^2 - 4}$ equal to zero, but only one of them is an x-intercept. Say which, and give the coordinates of the feature at the other value.

Answer: ________________

10. Jonah says: "$v(x) = \dfrac{x^2 - 4}{x - 2}$ has a vertical asymptote at $x = 2$, because the denominator is zero there." Explain why he is wrong, and describe exactly what the graph does at $x = 2$.

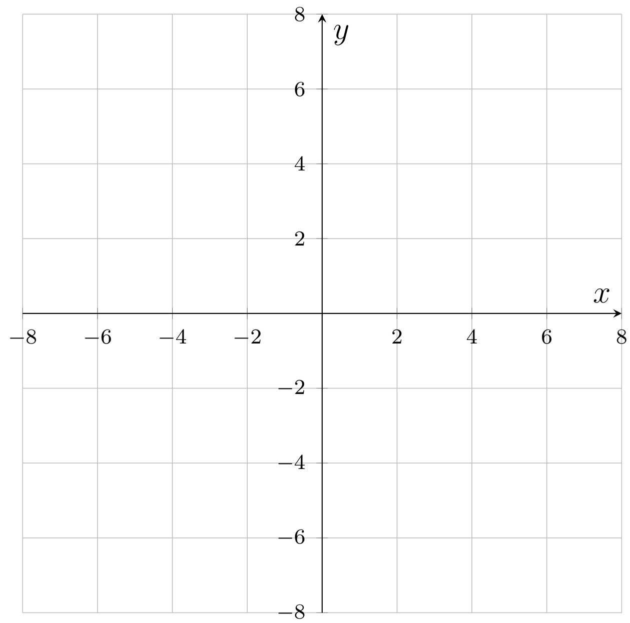
Answer: ________________

11. Let $w(x) = \dfrac{2x^2 + 3x}{x^2 - 9}$.

(a) Find the horizontal asymptote. (b) A graph may cross its horizontal asymptote. Does this one? If it does, find the x-value where it happens.

Answer: ________________

12. **Challenge.** Analyse $F(x) = \dfrac{2x^2 - 8}{x^2 - x - 2}$ completely: the hole (with coordinates), the vertical asymptote, the x-intercept, the y-intercept, and the horizontal asymptote. Then sketch the graph on the grid, marking the hole with an open circle and the asymptotes with dashed lines.



Answer: ____________________